

Stability Determination

UAV Design Project — Weekly Progress Summary

Skunk Works G-11

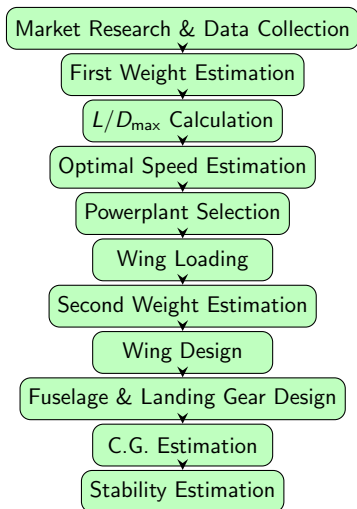
IIT Madras

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Overview

- 1 Overall Progress
- 2 Centre of Gravity Estimation
- 3 Longitudinal Static Stability
- 4 Key Results
- 5 Next Steps

Overall Design Progress



This Week's Focus

- ✓ C.G. Estimation (SolidWorks)
- ✓ Longitudinal Static Stability Analysis
- ✓ Wing & Fuselage Contributions
- ✓ Tail Sizing Requirements

Legend

■ Completed

Centre of Gravity Estimation — Method

Approach

- CAD-based mass property analysis in **SolidWorks**
- Full 3-D assembly model (fuselage, wing, etc.)
- Materials assigned via built-in library
- CG extracted using **Mass Properties** tool

Coordinate Convention

- Origin: **Nose** of the UAV
- X: positive toward nose
- Y: positive starboard
- Z: positive downward

CG Location (from nose)

Axis	Value
Longitudinal (X)	−241.06 mm
Lateral (Y)	≈ 0.00 mm
Vertical (Z)	9.22 mm

Lateral symmetry confirmed.

Longitudinal CG = 0.2411c from nose.

Mass Distribution of UAV Components

Component	Material	Density (kg/m ³)	Mass (kg)
Fuselage, Tail Boom	Al 6061, CF rod	2700, 1780	1.1466
Wing, Aileron	Al 6061, ABS	2700, 1020	1.3332
Motor	—	—	0.1450
Propeller	—	—	0.0480
Battery	—	—	1.5240
Landing Gear, Wheel	Al 6061, Rubber	2700, 1000	0.5650
Drop Payload	—	—	1.2000
Total			5.9618

Total UAV mass $\approx$ **5.96** kg — Battery & payload account for $\sim$ 46% of total mass

Longitudinal Static Stability — Conditions

Requirements for Stable Flight

- 1 Restoring pitching moment after disturbance
- 2 Negative pitching moment slope:

$$\frac{dC_m}{d\alpha} < 0$$

- 3 Trimmable at positive α :

$$C_{m_0} > 0$$

Current Status (Before Tail)

- Wing-body $C_{M_\alpha} > 0$
 $\Rightarrow$ Longitudinally **unstable**
- Net $C_M < 0$
 $\Rightarrow$ Nose-down tendency; **cannot trim**
- **Horizontal tail required** to:
 - Provide nose-up moment for trim
 - Make $C_{M_\alpha} < 0$

Wing Contribution to Pitching Moment

Governing Equation

$$C_{m_{cg,w}} = C_{m_{ac,w}} + (C_{L_{0,w}} + C_{L_{\alpha,w}} \alpha_w) \left(\frac{x_{cg}}{c} - \frac{x_{ac}}{c} \right)$$

Key Inputs (from XFLR5)

- Cruise $\alpha_w = 3^\circ$
- $C_L = 0.60$ (finite wing trim condition)
- $C_{L_{\alpha,w}} = 0.4935 / rad$
- $x_{ac} = 0.20376 \text{ m}$
- $x_{cg} = 0.24106 \text{ m}$ ($c = 0.2653 \text{ m}$)

Results

Quantity	Value
$C_{m_{Acw}}$ (XFLR5)	-0.2279
$C_{m_{L,w}}$	+0.08436
$C_{m_{cg,w}}$	-0.14354
$C_{m_{\alpha,w}}$	+0.06938 /rad

Wing alone: destabilizing slope ($C_{m_{\alpha}} > 0$), requires tail correction.

Wing Contribution (contd.)

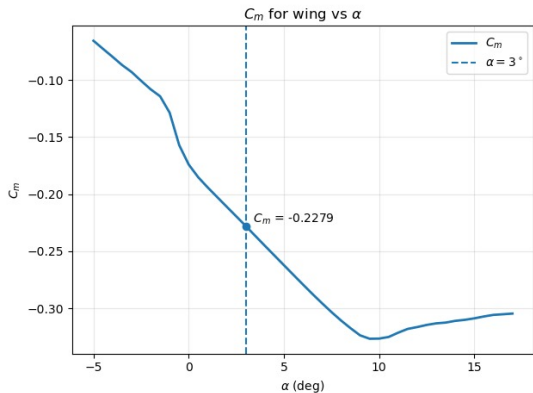


Figure: C_{m_w} v. α

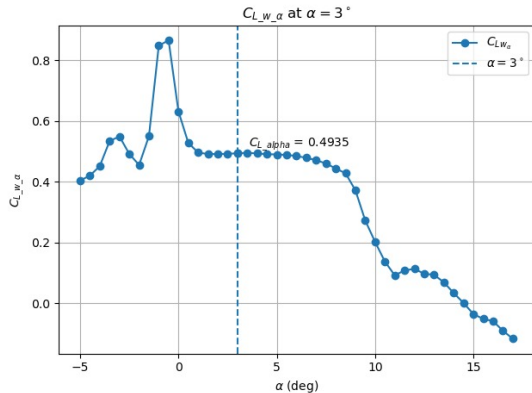


Figure: C_{L_α} v. α

Fuselage Contribution

Fuselage Geometry

- Width $b = 0.30$ m, Length $l = 0.80$ m
- Aspect ratio: $AR = b/l = 0.375$
- C_{L_α} (Helmhold equation):

$$C_{L_\alpha} = \frac{2\pi \times 0.375}{2 + \sqrt{4 + 0.375^2}} = 0.584 \text{ /rad}$$

Conclusion

- At $\alpha = 3^\circ$, fuselage $C_L \approx 0$
- No lift $\Rightarrow$ no lift-induced pitching moment
- Intrinsic moment also negligible
- **Fuselage contribution ≈ 0** in first analysis

Implication

The overall pitching moment is dominated by the **wing–tail interaction**.

Fuselage does **not** relieve the stability deficit.

Combined (Wing + Fuselage)

$$C_{m_\alpha} \approx +0.069 \text{ /rad}$$

Positive $\Rightarrow$ statically **unstable** without tail

Fuselage

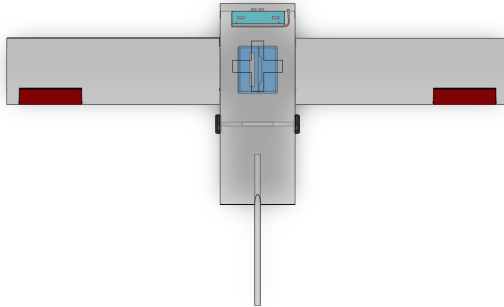


Figure: Components' distribution inside fuselage

Horizontal Tail — Role & Requirement

Tail Pitching Moment

$$C_{m_{\alpha,t}} = -\eta_t V_H C_{L_{\alpha,t}} \left(1 - \frac{d\varepsilon}{d\alpha} \right)$$

- $\eta_t \approx 0.9$ (tail efficiency)
- $V_H = S_t l_t / (S c)$ (tail volume ratio)
- $d\varepsilon/d\alpha \approx 0.3\text{--}0.5$ (downwash)

Dual Purpose of Tail

- ① **Trim:** generate nose-up moment (negative lift at tail) to achieve $C_M = 0$
- ② **Stability:** contribute negative $C_{M_{\alpha}}$ to satisfy $C_{M_{\alpha}} < 0$

Design Requirement

Tail must satisfy:

$$|C_{m_{\alpha,t}}| > C_{m_{\alpha,\text{wing+fus}}}$$

Appropriate tail **size**, **moment arm**, and **aerofoil** to be selected in the next iteration.

A tail that is too small $\Rightarrow$ marginally stable.
A tail that is too large $\Rightarrow$ overly stable, poor manoeuvrability.

Key Results Summary

Parameter	Value	Status
Total UAV Mass	5.962 kg	✓
CG Location (longitudinal)	-241.06 mm / 0.241c	✓
CG Location (lateral)	≈ 0 mm	✓ Symmetric
$C_{L_{\alpha,w}}$ (XFLR5)	0.4935 /rad	✓
Wing $C_{m_{cg,w}}$	-0.14354	✓ Destabilising slope
Fuselage $C_{m_{\alpha}}$	≈ 0	✓ Negligible
Combined $C_{m_{\alpha}}$ (w+f)	+0.069 /rad	✗ Unstable
Horizontal Tail	Required	→ Next Step

Next Steps

Immediate Tasks

- Horizontal tail sizing (area, span, chord)
- Select tail moment arm for adequate V_H
- Verify static margin:
 $SM = (x_{np} - x_{cg})/c > 0$
- Repeat XFLR5 analysis with full configuration

Open Issues

- Tail not yet included in CG estimate
- Downwash gradient $d\varepsilon/d\alpha$ to be refined
- Dynamic stability analysis pending

C.G. Estimation

Stability Estimation

Tail Sizing & Verification

Thank You

Questions?